AUTOSAR High-Level Design Specification (v2.0)

System Architecture: Body Control & Exterior Lighting Subsystem

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1. System Architectural Overview

This High-Level Design document specifies the software architecture for the Body Control Module (BCM). The BCM coordinates vehicle access, central door locking, exterior lighting control, and low-power standby modes. All software components adhere to the AUTOSAR component model communicating via the Virtual Function Bus (VFB) and Runtime Environment (RTE).

2. Software Components (SWC) Definition

The subsystem decomposes into the following architectural Software Components:

Component Name	SWC Type	Execution Period	Safety Category
BodyControlSWC	Application SWC	10 ms (Cyclic)	QM
DoorControlSWC	Sensor-Actuator SWC	20 ms (Cyclic)	ASIL-B
LightingSWC	Application SWC	20 ms (Cyclic - Modified)	ASIL-B
PowerDistributionSWC	Service SWC	100 ms (Event-driven)	ASIL-B
TelematicsGatewaySWC	Application SWC (New)	50 ms (Event-driven)	ASIL-B

3. Interface and Port Configuration

Software components interact strictly via AUTOSAR ports and interfaces configured in the AUTOSAR XML (ARXML) schema.

Interface Name	Interface Type	Data Elements / Operations	Cycle Time
DoorStatus_If	Sender-Receiver	DoorState (uint8), LockStatus (boolean), LatchSpeed (float32)	20 ms
LightControl_If	Client-Server	SetHeadlightState(uint8), GetFault()	Aperiodic
PowerMode_If	Mode-Switch	VehiclePowerState {OFF, ACC, RUN}	Event
RemoteCommand_If	Sender-Receiver (New)	RemoteLockReq (uint8), RemoteHazards (uint8)	10 ms

4. Port Connections and Dependencies

The following assembly connectors bind the required and provided ports across component boundaries:

• **Conn_DoorStatus**: Provided Port *DoorControlSWC.P_DoorStatus* connects to Required Port *BodyControlSWC.R_DoorStatus* using interface **DoorStatus_If**.

• **Conn_LightControl**: Provided Port *BodyControlSWC.P_LightControl* connects to Required Port *LightingSWC.R_LightControl* using interface **LightControl_If**.

• **Conn_PowerMode**: Provided Port *PowerDistributionSWC.P_PowerMode* broadcasts power mode changes to *BodyControlSWC.R_PowerMode*.

• **Conn_Telematics**: Provided Port *TelematicsGatewaySWC.P_RemoteCommand* connects to *BodyControlSWC.R_RemoteCommand* using interface **RemoteCommand_If**.

5. Functional Execution Flow & Timing Constraints

Scenario 1: Driver Door Unlock Sequence

1. The physical driver door handle capacitive sensor triggers an interrupt in the BSW I/O Hardware Abstraction.
2. The *DoorControlSWC* runnable *Run_ReadDoorSensors* executes and validates debounce timer (50 ms).
3. *DoorControlSWC* transmits *DoorStatus_If.DoorState* = *UNLOCKED* via sender-receiver port *P_DoorStatus*.
4. RTE delivers the signal to *BodyControlSWC*, activating runnable *Run_ProcessDoorEvents*.
5. *BodyControlSWC* commands *LightingSWC* via client-server call *SetHeadlightState(WELCOME_LIGHT)*.

6. Architectural Constraints & Diagnostics

Under ISO 26262 requirements, all inter-SWC communications crossing QM to ASIL boundaries must enforce memory partitioning and End-to-End (E2E) protection profiles. Specifically, *DoorControlSWC* (ASIL-B) transmission to *BodyControlSWC* (QM) must implement E2E Profile 01.